

Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

DDL, DML, DCL, TCL Commands

2520030098

Gutha Sushma Sree

Section - 3

Create the database:

```
CREATE DATABASE IF NOT EXISTS bookflow1_db;
```

Output:

```
✓ 88 15:52:08 CREATE DATABASE IF NOT EXISTS bookflow1_db
```

Use database:

```
USE bookflow1_db;
```

Output:

```
✓ 89 15:53:05 USE bookflow1_db
```

DDL (DATA DEFINITION LANGUAGE)

Create table:

```
CREATE TABLE bank_transactions (  
    txn_id INT PRIMARY KEY,  
    customer_name VARCHAR(50),  
    branch_name VARCHAR(50),  
    transaction_type VARCHAR(20),  
    amount DECIMAL(10,2),  
    transaction_date DATE  
);  
SELECT * FROM bank_transactions;
```

Output:

	txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
*	NULL	NULL	NULL	NULL	NULL	NULL

Alter table add column:

```
ALTER TABLE bank_transactions
ADD account_no VARCHAR(20);
SELECT * FROM bank_transactions;
```

Output:

	txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Alter table modify column

```
ALTER TABLE bank_transactions
MODIFY customer_name VARCHAR(100);
```

Output:

```
✓ 96 16:03:28 ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)
```

Rename table

```
RENAME TABLE bank_transactions
TO customer_transactions;
```

Output:

```
✓ 99 16:06:09 RENAME TABLE bank_transactions TO customer_transactions
```

Truncate table:

```
TRUNCATE TABLE customer_transactions;
```

Output:

```
✓ 57 19:04:52 TRUNCATE TABLE customer_transactions
```

Drop table

```
DROP TABLE IF EXISTS bank_backup;
```

Output:

```
⚠ 59 19:06:23 DROP TABLE IF EXISTS bank_backup
```

DML (DATA MANIPULATION LANGUAGE)

Create table:

```
CREATE TABLE bank_transactions (  
    txn_id INT PRIMARY KEY,  
    customer_name VARCHAR(50),  
    branch_name VARCHAR(50),  
    transaction_type VARCHAR(20),  
    amount DECIMAL(10,2),  
    transaction_date DATE  
);
```

Output:

	txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
*	NULL	NULL	NULL	NULL	NULL	NULL

Insert sample data:

```
INSERT INTO bank_transactions VALUES  
(101,'Sushma','Hyderabad','Deposit',5000,'2024-01-05'),  
(102,'Sowmya','Hyderabad','Withdrawal',2000,'2024-01-06'),  
(103,'Mokshagna','Vijayawada','Deposit',12000,'2024-01-08'),  
(104,'Venugopal','Vizag','Deposit',8000,'2024-01-10'),  
(105,'Kalavathi','Hyderabad','Withdrawal',3500,'2024-01-11'),  
(106,'Govinda','Vizag','Deposit',15000,'2024-01-12'),  
(107,'Krishna','Vijayawada','Withdrawal',1000,'2024-01-13'),  
(108,'Sairam','Hyderabad','Deposit',9000,'2024-01-14'),  
(109,'Laxmi','Vizag','Withdrawal',4000,'2024-01-15'),  
(110,'Saraswati','Vijayawada','Deposit',11000,'2024-01-16');  
SELECT * FROM bank_transactions;
```

Output:

	txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
▶	101	Sushma	Hyderabad	Deposit	5000.00	2024-01-05
	102	Sowmya	Hyderabad	Withdrawal	2000.00	2024-01-06
	103	Mokshagna	Vijayawada	Deposit	12000.00	2024-01-08
	104	Venugopal	Vizag	Deposit	8000.00	2024-01-10
	105	Kalavathi	Hyderabad	Withdrawal	3500.00	2024-01-11
	106	Govinda	Vizag	Deposit	15000.00	2024-01-12
	107	Krishna	Vijayawada	Withdrawal	1000.00	2024-01-13
	108	Sairam	Hyderabad	Deposit	9000.00	2024-01-14
	109	Laxmi	Vizag	Withdrawal	4000.00	2024-01-15
	110	Saraswati	Vijayawada	Deposit	11000.00	2024-01-16
*	NULL	NULL	NULL	NULL	NULL	NULL

Insert

```
INSERT INTO bank_transactions
VALUES
(111,'Venu','Vizag','Deposit',7000,'2024-01-18');
SELECT * FROM bank_transactions;
```

Output:

	txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
▶	101	Sushma	Hyderabad	Deposit	5000.00	2024-01-05
	102	Sowmya	Hyderabad	Withdrawal	2000.00	2024-01-06
	103	Mokshagna	Vijayawada	Deposit	12000.00	2024-01-08
	104	Venugopal	Vizag	Deposit	8000.00	2024-01-10
	105	Kalavathi	Hyderabad	Withdrawal	5000.00	2024-01-11
	106	Govinda	Vizag	Deposit	15000.00	2024-01-12
	107	Krishna	Vijayawada	Withdrawal	1000.00	2024-01-13
	108	Sairam	Hyderabad	Deposit	9000.00	2024-01-14
	109	Laxmi	Vizag	Withdrawal	4000.00	2024-01-15
	110	Saraswati	Vijayawada	Deposit	11000.00	2024-01-16
	111	Venu	Vizag	Deposit	7000.00	2024-01-18

Update:

```
UPDATE bank_transactions
SET amount = 5000
WHERE txn_id = 105;
SELECT * FROM bank_transactions;
```

Output:

	txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
▶	101	Sushma	Hyderabad	Deposit	5000.00	2024-01-05
	102	Sowmya	Hyderabad	Withdrawal	2000.00	2024-01-06
	103	Mokshagna	Vijayawada	Deposit	12000.00	2024-01-08
	104	Venugopal	Vizag	Deposit	8000.00	2024-01-10
	105	Kalavathi	Hyderabad	Withdrawal	5000.00	2024-01-11
	106	Govinda	Vizag	Deposit	15000.00	2024-01-12
	107	Krishna	Vijayawada	Withdrawal	1000.00	2024-01-13
	108	Sairam	Hyderabad	Deposit	9000.00	2024-01-14
	109	Laxmi	Vizag	Withdrawal	4000.00	2024-01-15
	110	Saraswati	Vijayawada	Deposit	11000.00	2024-01-16
	111	Venu	Vizag	Deposit	7000.00	2024-01-18

Delete & Select

```
DELETE FROM bank_transactions
WHERE txn_id = 111;
SELECT * FROM bank_transactions;
```

Output:

	txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
▶	101	Sushma	Hyderabad	Deposit	5000.00	2024-01-05
	102	Sowmya	Hyderabad	Withdrawal	2000.00	2024-01-06
	103	Mokshagna	Vijayawada	Deposit	12000.00	2024-01-08
	104	Venugopal	Vizag	Deposit	8000.00	2024-01-10
	105	Kalavathi	Hyderabad	Withdrawal	5000.00	2024-01-11
	106	Govinda	Vizag	Deposit	15000.00	2024-01-12
	107	Krishna	Vijayawada	Withdrawal	1000.00	2024-01-13
	108	Sairam	Hyderabad	Deposit	9000.00	2024-01-14
	109	Laxmi	Vizag	Withdrawal	4000.00	2024-01-15
	110	Saraswati	Vijayawada	Deposit	11000.00	2024-01-16
*	NULL	NULL	NULL	NULL	NULL	NULL

Select with where

```
SELECT *
FROM bank_transactions
WHERE transaction_type='Deposit';
```

Output:

	txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
▶	101	Sushma	Hyderabad	Deposit	5000.00	2024-01-05
	103	Mokshagna	Vijayawada	Deposit	12000.00	2024-01-08
	104	Venugopal	Vizag	Deposit	8000.00	2024-01-10
	106	Govinda	Vizag	Deposit	15000.00	2024-01-12
	108	Sairam	Hyderabad	Deposit	9000.00	2024-01-14
	110	Saraswati	Vijayawada	Deposit	11000.00	2024-01-16
*	NULL	NULL	NULL	NULL	NULL	NULL

Select with order by

```
SELECT *
FROM bank_transactions
ORDER BY amount DESC;
```

Output:

	txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
▶	106	Govinda	Vizag	Deposit	15000.00	2024-01-12
	103	Mokshagna	Vijayawada	Deposit	12000.00	2024-01-08
	110	Saraswati	Vijayawada	Deposit	11000.00	2024-01-16
	108	Sairam	Hyderabad	Deposit	9000.00	2024-01-14
	104	Venugopal	Vizag	Deposit	8000.00	2024-01-10
	101	Sushma	Hyderabad	Deposit	5000.00	2024-01-05
	105	Kalavathi	Hyderabad	Withdrawal	5000.00	2024-01-11
	109	Laxmi	Vizag	Withdrawal	4000.00	2024-01-15
	102	Sowmya	Hyderabad	Withdrawal	2000.00	2024-01-06
	107	Krishna	Vijayawada	Withdrawal	1000.00	2024-01-13
*	NULL	NULL	NULL	NULL	NULL	NULL

DCL (DATA CONTROL LANGUAGE)

Grant all privileges

```
CREATE USER 'BranchManager'@'localhost'  
IDENTIFIED BY 'Branch@123';  
GRANT ALL PRIVILEGES  
ON bookflow1_db.bank_transactions  
TO 'BranchManager'@'localhost';  
SHOW GRANTS FOR 'BranchManager'@'localhost';
```

Output:

```
✓ 97 22:36:21 SHOW GRANTS FOR 'BranchManager'@'localhost'
```

Revoke all

```
REVOKE ALL PRIVILEGES  
ON bookflow1_db.bank_transactions  
FROM 'BranchManager'@'localhost';
```

Output:

```
✓ 98 22:38:01 REVOKE ALL PRIVILEGES ON bookflow1_db.bank_transactions FROM 'BranchManager'@'localhost'
```

TCL (TRANSACTION CONTROL LANGUAGE)

Commit

```
UPDATE bank_transactions  
SET amount=6000  
WHERE txn_id=101;  
COMMIT;
```

Output:

```
✓ 105 22:45:10 UPDATE bank_transactions SET amount=6000 WHERE txn_id=101  
✓ 106 22:45:12 COMMIT
```

Savepoint:

```
SAVEPOINT Before_Update;
```

Output:

```
✓ 107 22:46:42 SAVEPOINT Before_Update
```

Rollback To Savepoint

```
UPDATE bank_transactions
SET amount=99999
WHERE txn_id=102;
```

```
SELECT * FROM bank_transactions;
```

Output:

	txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
▶	101	Sushma	Hyderabad	Deposit	6000.00	2024-01-05
	102	Sowmya	Hyderabad	Withdrawal	99999.00	2024-01-06
	103	Mokshagna	Vijayawada	Deposit	12000.00	2024-01-08
	104	Venugopal	Vizag	Deposit	8000.00	2024-01-10
	105	Kalavathi	Hyderabad	Withdrawal	5000.00	2024-01-11
	106	Govinda	Vizag	Deposit	15000.00	2024-01-12
	107	Krishna	Vijayawada	Withdrawal	1000.00	2024-01-13
	108	Sairam	Hyderabad	Deposit	9000.00	2024-01-14
	109	Laxmi	Vizag	Withdrawal	4000.00	2024-01-15
	110	Saraswati	Vijayawada	Deposit	11000.00	2024-01-16
•	NULL	NULL	NULL	NULL	NULL	2024-01-